Probability Is All You Need

- History
- Descriptive Statistics
- Definition of Probability
- Probability and Distribution, Expectation, Variance
- Probability and Central Limit Theorem
- Probability and Hypothesis Test
- Probability and Bayes
- Probability and Stochastic Process

History

The birth of statistics

- Math: Pascal and Fermat's researches on gambling. Since 1654.
- Physics: Einstein, Perrin, Langevin, Ito's researches on Brownian Motion. Since 1905.
- Biology: Galton, Pearson, Fisher's researches on Eugenics. Since 1883.
- NLP: Markov's researches on Pushkin's poem *Eugene Onegin*. Since 1913.

In the math part, Bernouli, De Moivre, Laplace, Poisson also made great contributions. In 1900, Hilbert's sixth question is about the axiomatization of probability and physics. In 1933, Kolmogorov did that.

But the true birth of statistics relied on the latter 3 parts, especially biology. **From the birth, statistics has deep connections with biology, physics and NLP.**

Descriptive Statistics

In primary school, junior high school and senior high school, we learnt a lot about the so-called "descriptive statistics".

For example

- boy-girl-ratio: France. China.
    - In 1814, Laplace wrote in "A Philosophical Essay on Probabilities" that the ratio of male to female infants in London, Berlin, and throughout France was 104.76:100, while in Paris it was 104.17:100.
- basketball, soccer: see kaggle.
- weighted average: like grades weighted by points.

But we can't just do this kind of thing, we must go beyond, build a model. Sometimes people call this "statistical inference". Here are 3 important things we need to do.

- population: just like the population of a country. (Your Model)

- sample: both a number and a random variable. (Your Data)
- f -> p

Definition of Probability

- Just like throwing a coin
- Measure

Probability and Distribution, Expectation, Variance

- Distribution: Discrete and Continuous
- E, Var: See them as Operator!
  - $E(b) = E_x(E_b(b|x))$
  - $Var(b) = E_x(Var_b(b|x)) + Var_x(E_b(b|x))$
  - MSE = Bias + Variance + Noise
  - $E(\frac{1}{n-1}\sum_{i=1}^{n}(X_i - mean\ of\ the\ data)^2) = Var(X)$

Example from RL: remember the slot machine in class?

I must emphasize that the distribution has more information than E, Var. That means, when you get a data in your project, first plot its distribution!

Probability and Central Limit Theorem

Wikipedia.

In the simplest form,

$$\forall i, X_i\ i.i.d. \Rightarrow \sum_{i=1}^{n} X_i \xrightarrow{d} N(nE(X_i), nVar(X_i))$$

And this is the true reason why Gauss choose e^{-x^2} instead of $cos(x), e^{-x^4}, (\frac{1}{1+e^{-x}})', (arctan(x))'$ and so on.

There is an example from neuroscience, which I will call the core of biological neural network. Based on Central Limit Theorem, a neuron in a big neural network should have $\frac{\sigma}{\mu} \to 0, as\ n \to \infty$, but this contradicts with the experiment result, which show finite $\frac{\sigma}{\mu}$.

There are 2 solutions

- balanced exhibition and inhibition
- correlated inputs

Probability and Hypothesis Test

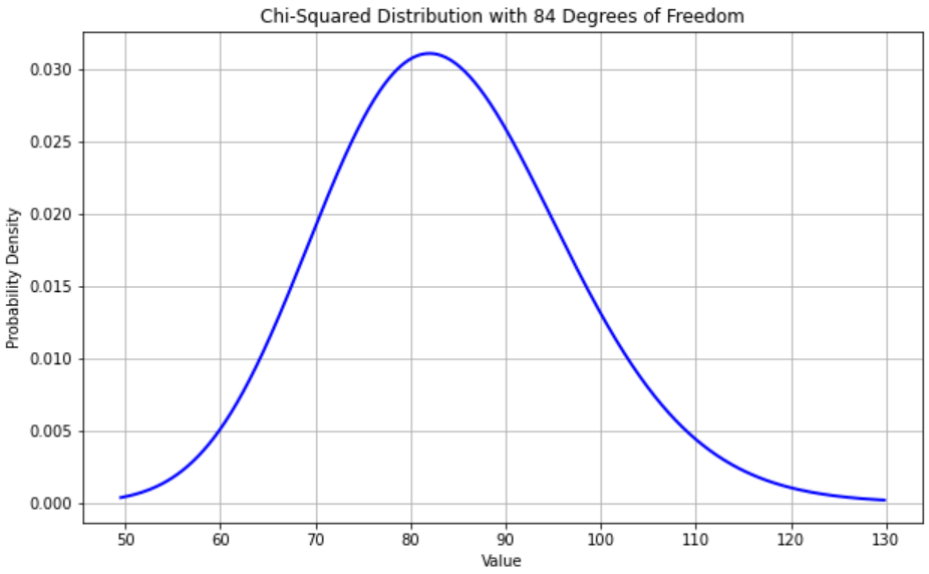
Lady Tasting Tea

Other hypothesis test: χ^2, t, F, permutation test, rank, KS, ...

Further Reading

- Do you believe that certain zodiac signs are more compatible with each other?
- The Misuse of p-value in the Senior High School
- p-hacking: What it is and how to avoid it!
- Example from Neuroscience: Sen Song's work

Fisher vs Mendel



Common Hypothesis Tests

表 6.2.1: 一样本正态总体$N(\mu, \sigma^2)$.

检验对象	检验统计量	分布	拒绝域†
μ (σ^2已知)	$U = \sqrt{n}(\bar{X} - \mu_0)/\sigma$	$N(0,1)$	$\begin{cases} \lvert U\rvert > u_{\alpha/2} \\ U > u_\alpha \\ U < u_\alpha \end{cases}$
μ (σ^2 未知)	$T = \sqrt{n}(\bar{X} - \mu_0)/S$	t_{n-1}	$\begin{cases} \lvert T\rvert > t_{n-1}(\alpha/2) \\ T > t_{n-1}(\alpha) \\ T < t_{n-1}(\alpha) \end{cases}$
σ^2 (μ 已知)	$\chi^2 = \frac{1}{\sigma_0^2}\sum_{i=1}^n (X_i - \mu)^2$	χ_n^2	$\begin{cases} \chi^2 > \chi_n^2(\alpha/2)\text{或者}\chi^2 < \chi_n^2(1-\alpha/2) \\ \chi^2 > \chi_n^2(\alpha) \\ \chi^2 < \chi_n^2(1-\alpha) \end{cases}$
σ^2 (μ 未知)	$\chi^2 = \frac{1}{\sigma_0^2}\sum_{i=1}^n (X_i - \bar{X})^2$	χ_{n-1}^2	$\begin{cases} \chi^2 > \chi_{n-1}^2(\alpha/2)\text{或者}\chi^2 < \chi_{n-1}^2(1-\alpha/2) \\ \chi^2 > \chi_{n-1}^2(\alpha) \\ \chi^2 < \chi_{n-1}^2(1-\alpha) \end{cases}$

†有关均值的检验：对立假设分别为$\mu \neq \mu_0$, $\mu > \mu_0$ 和$\mu < \mu_0$. 有关方差的检验：对立假设分别为$\sigma^2 \neq \sigma_0^2$, $\sigma^2 > \sigma_0^2$ 和$\sigma^2 < \sigma_0^2$.

表 6.2.2: 两样本正态总体的假设检验

检验对象	检验统计量	分布	拒绝域†
均值(方差已知)	$U=\frac{\bar{X}-\bar{Y}}{\sqrt{\frac{\sigma_1^2}{m}+\frac{\sigma_2^2}{n}}}$	$N(0,1)$	$\begin{cases}\lvert U\rvert>u(\alpha/2)\\ U>u(\alpha)\\ U<-u(\alpha)\end{cases}$
均值(方差未知)‡	$T=\frac{\bar{X}-\bar{Y}}{S_w\sqrt{\frac{1}{m}+\frac{1}{n}}}$	t_{m+n-2}	$\begin{cases}\lvert T\rvert>t_{m+n-2}(\alpha/2)\\ T>t_{m+n-2}(\alpha)\\ T<-t_{m+n-2}(\alpha)\end{cases}$
方差(均值已知)	$F=\frac{\sum_{i=1}^m(X_i-\mu_1)^2/m}{\sum_{i=1}^n(X_i-\mu_2)^2/n}$	$F_{m,n}$	$\begin{cases}F>F_{m,n}(\alpha/2)\text{或}F<\frac{1}{F_{n,m}(\alpha/2)}\\ F>F_{m,n}(\alpha)\\ F<\frac{1}{F_{n,m}(\alpha)}\end{cases}$
方差(均值未知)	$F=\frac{S_1^2}{S_2^2}$	$F_{m-1,n-1}$	$\begin{cases}F>F_{m-1,n-1}(\alpha/2)\text{或}F<\frac{1}{F_{n-1,m-1}(\alpha/2)}\\ F>F_{m-1,n-1}(\alpha)\\ F<\frac{1}{F_{n-1,m-1}(\alpha)}\end{cases}$

†有关均值的检验：对立假设分别为$\mu_1\neq\mu_2$, $\mu_1>\mu_2$ 和$\mu_1<\mu_2$. 有关方差的检验：对立假设分别为$\sigma_1^2\neq\sigma_2^2$, $\sigma_1^2>\sigma_2^2$ 和$\sigma_1^2<\sigma_2^2$.

‡假定方差相等

Probability and Bayes

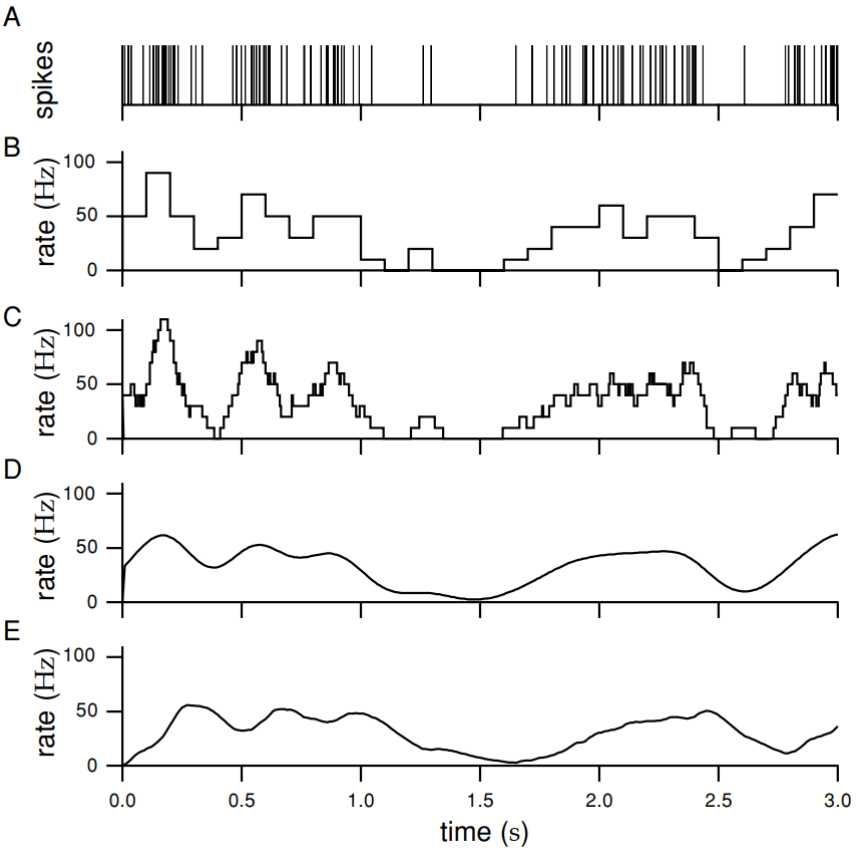
- Hypothesis Test
- Three door problem
- Three choice problem

Probability and Stochastic Process

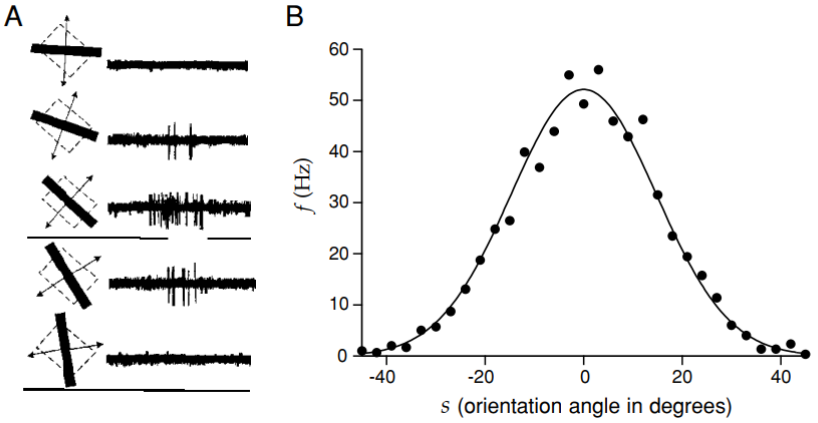
- Markov Process
- Poisson Process
- Steady Process

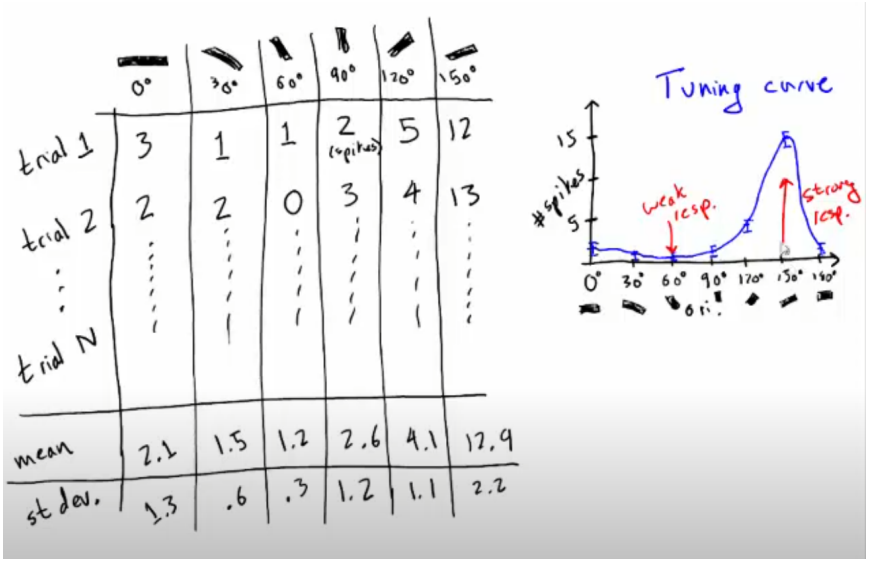
Stochastic Process and Neuroscience

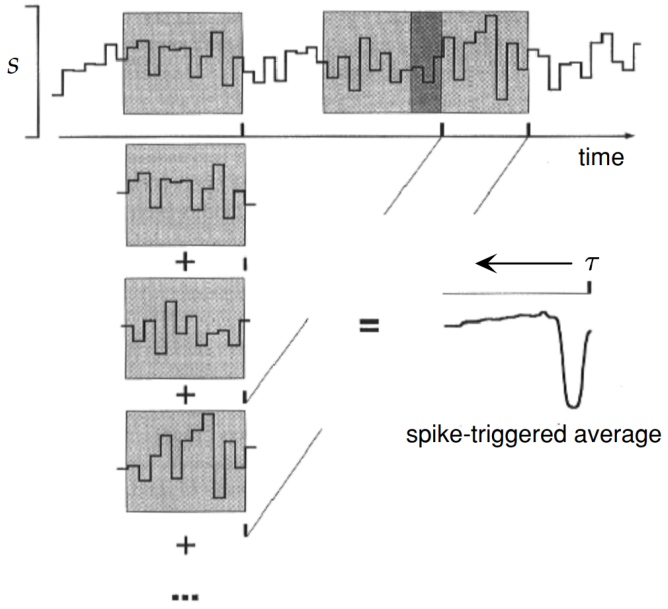
Count



Error-Bar





(figure from the Internet)

The-stimulus-before-spikes